8D Report - Rotor Shaft Bearing Seat Out of Tolerance

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D1 Team

Team leader: quality engineer machining. Members: machining process engineer, maintenance technician MC-500, metrology lab.

D2 Problem description

During the final inspection on 2026-09-03, 38 rotor shafts RS-40 from machine MC-504 had a bearing seat diameter below the lower limit of 40.002 mm (measured 39.996 to 40.001 mm). The parts were produced in the early shift after the machine was down for the weekend.

D3 Containment actions

- All shafts from MC-504 since the last good measurement were blocked (212 parts) and 100 % measured. 38 parts were scrapped.
- The in-process measurement frequency for the bearing seat on all MC-500 machines was raised from every 25th part to every 10th part until the 8D is closed.

D4 Root cause (preliminary)

The spindle warm-up program O9001 was not run after the weekend standstill. The thermal growth of the spindle during the first hour shifted the diameter. The warm-up step is written in MM-CNC-003, but WI-MCH-022 only refers to it and the operators were not aware it is mandatory.

D5 Planned corrective actions

1. Interlock in the machine control: production program cannot start after more than 8 hours of standstill until O9001 has been run. Due: 2026-11-15.
2. Add the warm-up as a numbered step in WI-MCH-022 Rev B. Due: 2026-10-31.
3. Training of all MC-500 operators. Due: 2026-10-31.

D6 Status

Open. Actions 2 and 3 are in progress. Scrap cost so far: 4,940 EUR.